

Strategy Brief #7

Groundwater Management in Data-Scarce Urban Regions



This examines how the absence of monitoring systems and governance frameworks is accelerating groundwater depletion, land subsidence, and water insecurity in rapidly urbanizing cities, and what decision-makers must do to secure long-term water sustainability.

Gergino Chounna Y.

<https://chounnagerginoy.github.io/>

Introduction

Groundwater is a critical resource in rapidly urbanizing regions, often used without adequate monitoring or regulation.



Key Challenge

Over-extraction and poor management lead to **resource depletion, subsidence, and water insecurity.**

Strategic Insights

- Groundwater is often **invisible and poorly governed.**
- Data scarcity limits **effective management and planning.**
- Urban demand is increasing **faster than supply systems.**



Strategic Implications

- Risk of **water crises in major cities.**
- Increased inequality in water access.
- Long-term environmental degradation.



Recommendations

- Develop **groundwater monitoring systems**.
- Strengthen **governance and regulation frameworks**.
- Promote **sustainable water use practices**.
- Integrate groundwater into **urban planning strategies**.



Value Proposition

Sustainable groundwater management ensures **long-term water security and urban resilience**.